

Practice Quiz: Perception

Part A: Multiple Choice

1. In the predictive coding architecture, top-down predictions are sent by: A) Superficial pyramidal cells in layers 2/3 B) Deep pyramidal cells in layers 5/6 C) Inhibitory interneurons in layer 1 D) Granular cells in layer 4
2. Sensory attenuation refers to: A) The gradual loss of hearing with age B) Reduced perception of self-generated stimuli due to efference copy-based prediction C) The brain's inability to process two sensory modalities simultaneously D) A form of amnesia affecting sensory memories
3. The inability to tickle yourself is evidence for: A) The separation of motor and sensory systems B) Active Inference — the brain predicts and cancels expected sensory consequences of its own actions C) The superiority of touch over other senses D) Social bonding mechanisms
4. Hallucinations in the Active Inference framework arise from: A) Damage to the retina or cochlea B) High-precision top-down predictions that override or substitute for bottom-up sensory evidence C) Complete absence of neural activity in sensory cortex D) Excessive attention to environmental stimuli
5. Mismatch negativity (MMN) is considered a neural signature of: A) Motor planning B) Prediction error in response to violations of expected auditory patterns C) Emotional processing of surprising events D) Memory retrieval
6. Charles Bonnet syndrome hallucinations occur because: A) The patient has a psychiatric disorder B) Reduced sensory input (visual loss) allows top-down predictions to dominate without bottom-up constraint C) The visual cortex is damaged D) The patient is imagining visual experiences intentionally
7. Failed sensory attenuation in schizophrenia may lead to auditory hallucinations because: A) Inner speech is not predicted as self-generated and is therefore experienced as

coming from an external source B) The auditory cortex is over-developed C) The patient has hearing loss D) Social isolation reduces the need for inner speech

Part B: Short Answer

1. Explain how binocular rivalry (different images presented to each eye producing alternating perception) can be interpreted as competition between top-down predictions. What role does precision play?
2. Patient hears voices commenting on their actions. They are cognitively intact but convinced the voices are external. Analyze this as failed sensory attenuation: what specific mechanism has broken down?
3. Predict what would happen to Charles Bonnet syndrome hallucinations if the patient's visual input were artificially increased (e.g., through a high-contrast display). Justify your prediction using Active Inference principles.